

CHAPTER 8

Cell: The Unit of Life

An Overview of Cell and Cell Theory

- The concept of “*Omnis cellula-e cellula*” regarding cell division was first proposed by (2019)
 - Rudolf Virchow
 - Theodor Schwann
 - Schleiden
 - Aristotle
- Cellular organelles with membranes are: (2015 Re)
 - Chromosomes, ribosomes and endoplasmic reticulum
 - Endoplasmic reticulum, ribosomes and nuclei
 - Lysosomes, Golgi apparatus and mitochondria
 - Nuclei, ribosome and mitochondria
- Which of the following structures is not found in a prokaryotic cell? (2015 Re)
 - Ribosome
 - Mesosome
 - Plasma membrane
 - Nuclear envelope
- Which of the following is not membrane-bound? (2015 Re)
 - Ribosomes
 - Lysosomes
 - Mesosomes
 - Vacuoles

Prokaryotic Cells

- Mesosome in a cell is a: (2024 Re)
 - membrane bound vesicular structure.
 - chain of many ribosomes attached to a single mRNA.
 - special structure formed by extension of plasma membrane.
 - medium sized chromosome.
- Given below are two statements: (2023-Manipur)

Statement-I: In bacteria, the mesosomes are formed by the extensions of plasma membrane.

Statement-II: The mesosomes, in bacteria, help in DNA replication and cell wall formation.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

 - Statement-I** is correct but **Statement-II** is incorrect.
 - Statement-I** is incorrect but **Statement-II** is correct.
 - Both **Statement-I** and **Statement-II** are correct.
 - Both **Statement-I** and **Statement-II** are incorrect.
- Given below are two statements: (2022)

Statement-I: Mycoplasma can pass through less than 1 micron filter size.

Statement-II: Mycoplasma are bacteria with cell wall

In the light of the above statements, choose the most appropriate answer from the options given below.

- Statement-I is incorrect but Statement-II is correct
 - Both Statement-I and Statement-II are correct
 - Both statement-I and statement-II are incorrect
 - Statement-I is correct but Statement-II is incorrect
- Which of the following statements about inclusion bodies is incorrect? (2020)
 - These are involved in ingestion of food particles.
 - They lie free in the cytoplasm
 - These represent reserve material in cytoplasm
 - They are not bound by any membrane
 - Inclusion bodies of blue-green, purple and green photosynthetic bacteria are: (2020-Covid)
 - Gas vacuoles
 - Centrioles
 - Microtubules
 - Contractile vacuoles
 - The size of Pleuropneumonia - like Organism (PPLO) is: (2020-Covid)
 - 1 - 2 μm
 - 10 - 20 μm
 - 0.1 μm
 - 0.02 μm
 - Which among the following is not a prokaryote? (2018)
 - Saccharomyces*
 - Mycobacterium*
 - Nostoc*
 - Oscillatoria*
 - Many ribosomes may associate with a single mRNA to form multiple copies of a polypeptide simultaneously. Such strings of ribosomes are termed as (2018)
 - Polysome
 - Polyhedral bodies
 - Plastidome
 - Nucleosome
 - Which of the following components provides sticky character to the bacterial cell? (2017-Delhi)
 - Cell wall
 - Nuclear membrane
 - Plasma membrane
 - Glycocalyx
 - Select the wrong statement: (2016 - II)
 - Cyanobacteria lack flagellated cells.
 - Mycoplasma is a wall-less microorganism
 - Bacterial cell wall is made up of peptidoglycan.
 - Pili and fimbriae are mainly involved in motility of bacterial cells
 - Select the mismatch: (2016 - II)
 - Protists-Eukaryotes
 - Methanogens-Prokaryotes
 - Gas vacuoles-Green bacteria
 - Large central vacuoles-Animal cells

16. A complex of ribosomes attached to a single strand of RNA is known: (2016 - I)
 a. Polysome b. Polymer
 c. Polypeptide d. Okazaki fragment
17. Which one of the following is not an inclusion body found in prokaryotes? (2015)
 a. Glycogen granule b. Polysome
 c. Phosphate granule d. Cyanophycan granule
18. Chromatophores take part in: (2015 Re)
 a. Growth b. Movement
 c. Respiration d. Photosynthesis
19. The structures that help some bacteria to attach to rocks and or host tissues are: (2015 Re)
 a. Fimbriae b. Mesosomes
 c. Holdfast d. Rhizoids
20. The motile bacteria are able to move by: (2014)
 a. Pili b. Fimbriae c. Flagella d. Cilia

Endomembrane System

21. Given below are two statements: (2024 Re)
Statement-I: Concentrically arranged cisternae of Golgi complex are arranged near the nucleus with distinct convex *cis* or maturing and concave *trans* or forming face.
Statement-II: A number of proteins are modified in the cisternae of Golgi complex before they are released from *cis* face.
 In the light of the above statements, choose the correct answer from the options given below.
 a. Statement-I is true but Statement-II is false.
 b. Statement-I is false but Statement-II is true.
 c. Both Statement-I and Statement-II are true.
 d. Both Statement-I and Statement-II are false
22. Which of the following are NOT considered as the part of endomembrane system? (2023)
 A. Mitochondria B. Endoplasmic Reticulum
 C. Chloroplasts D. Golgi complex
 E. Peroxisomes
 Choose the most appropriate answer from the options given below:
 a. A, D and E only b. B and D only
 c. A, C and E only d. A and D only
23. Which of the following statements are **correct** with respect of Golgi apparatus? (2023-Manipur)
 A. It is the important site of formation of glycoprotein and glycolipids.
 B. It produces cellular energy in the form of ATP.
 C. It modifies the protein synthesized by ribosomes on ER.
 D. It facilitates the transport of ions.
 E. It provides mechanical support.
 Choose the **most appropriate** answer from the options given below:
 a. (B) and (C) only b. (A) and (C) only
 c. (A) and (D) only d. (D) and (E) only
24. If the pH in lysosomes is increased to alkaline, what will be the outcome? (2022 Re)
 a. Lysosomal enzymes will be more active
 b. Hydrolytic enzymes will function more efficiently
 c. Hydrolytic enzymes will become inactive
 d. Lysosomal enzymes will be released into the cytoplasm
25. The organelles that are included in the endomembrane system are: (2021)
 a. Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles.
 b. Golgi complex, Mitochondria, Ribosomes and Lysosomes.
 c. Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes.
 d. Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes.
26. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells? (2020)
 a. Peroxisomes b. Golgi bodies
 c. Polysomes d. Endoplasmic reticulum
27. Which of the following statements is **not** correct? (2019)
 a. Lysosomes have numerous hydrolytic enzymes.
 b. The hydrolytic enzymes of lysosomes are active under acidic pH.
 c. Lysosomes are membrane bound structures.
 d. Lysosomes are formed by the process of packaging in the endoplasmic reticulum.
28. The Golgi complex participates in (2018)
 a. Fatty acid breakdown
 b. Formation of secretory vesicles
 c. Respiration in bacteria
 d. Activation of amino acid
29. Which of the following events does not occur in rough endoplasmic reticulum? (2018)
 a. Protein folding
 b. Protein glycosylation
 c. Cleavage of signal peptide
 d. Phospholipid synthesis
30. A cell organelle containing hydrolytic enzymes is: (2016 - II)
 a. Ribosome b. Mesosome
 c. Lysosome d. Microsome
31. Select the correct matching in the following pairs: (2015)
 a. Rough ER – Synthesis of glycogen
 b. Rough ER – Oxidation of fatty acids
 c. Smooth ER – Oxidation of phospholipids
 d. Smooth ER – Synthesis of lipids
32. The osmotic expansion of a cell kept in water is chiefly regulated by: (2014)
 a. Ribosomes b. Mitochondria
 c. Vacuoles d. Plastids

33. Which one of the following organelle in the figure correctly matches with its function? (2013)



- a. Rough endoplasmic reticulum, protein synthesis
b. Rough endoplasmic reticulum, formation of glycoproteins
c. Golgi apparatus, protein synthesis
d. Golgi apparatus, formation of glycolipids
34. A major site for synthesis of lipids is: (2013)
a. Nucleoplasm b. RER
c. SER d. Symplast
35. The Golgi complex plays a major role: (2013)
a. In post translational modification of proteins and glycosidation of lipids
b. In trapping the light and transforming it into chemical energy
c. In digesting proteins and carbohydrates
d. As energy transferring organelles

Mitochondria, Plastids, Ribosomes

36. The DNA present in chloroplast is: (2024)
a. Linear, single stranded b. Circular, single stranded
c. Linear, double stranded d. Circular, double stranded
37. Given below are two statements: (2024)
Statement-I: Mitochondria and chloroplasts are both double membrane bound organelles.
Statement-II: Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.
In the light of the above statements, choose the most appropriate answer from the options given below:
a. Statement-I is correct but Statement-II is incorrect.
b. Statement-I is incorrect but Statement-II is correct.
c. Both Statement-I and Statement-II are correct.
d. Both Statement-I and Statement-II are incorrect.
38. Which of the following pair of organelles does **not** contain DNA? (2019)
a. Mitochondria and Lysosomes
b. Chloroplast and Vacuoles
c. Lysosomes and Vacuoles
d. Nuclear envelope and Mitochondria
39. Which of the following statements regarding mitochondria is incorrect? (2019)
a. Outer membrane is permeable to monomers of carbohydrates, fats and proteins.
b. Enzymes of electron transport are embedded in outer membrane.
c. Inner membrane is convoluted with infoldings.
d. Mitochondrial matrix contains single circular DNA molecule and ribosomes.

40. Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP? (2017-Delhi)

a. Lysosome b. Ribosome
c. Chloroplast d. Mitochondrion

41. Mitochondria and chloroplast are
A. Semi-autonomous organelles
B. Formed by division of pre-existing organelles and they contain DNA but lack protein synthesizing machinery
Which one of the following options is correct? (2016-I)

a. Both (A) and (B) are correct
b. (B) is true but (A) is false
c. (A) is true but (B) is false
d. Both (A) and (B) are false

42. The structures that are formed by stacking of organized flattened membranous sacs in the chloroplasts are: (2015)

a. Stroma lamellae b. Stroma
c. Cristae d. Grana

43. DNA is not present in: (2015)

a. Nucleus b. Mitochondria
c. Chloroplast d. Ribosomes

44. Which structures perform the function of mitochondria in bacteria? (2014)

a. Mesosomes b. Nucleoid
c. Ribosomes d. Cell wall

Cytoskeleton, Cilia and Flagella, Centrosome and Centrioles

45. Which of the following functions is carried out by cytoskeleton in a cell? (2023)

a. Transportation b. Nuclear division
c. Protein synthesis d. Motility

46. Match the following columns and select the correct option: (2020-Covid)

Column-I		Column-II	
A.	Smooth Endoplasmic Reticulum	i.	Protein synthesis
B.	Rough endoplasmic reticulum	ii.	Lipid synthesis
C.	Golgi complex	iii.	Glycosylation
D.	Centriole	iv.	Spindle formation

A	B	C	D	A	B	C	D		
a.	(iii)	(i)	(ii)	(iv)	b.	(i)	(ii)	(iii)	(iv)
c.	(iv)	(ii)	(i)	(iii)	d.	(ii)	(i)	(iii)	(iv)

47. Microtubules are the constituents of: (2016-I)

a. Cilia, Flagella and Peroxisomes
b. Spindle fibres, Centrioles and Cilia
c. Centrioles, Spindle fibres and Chromatin
d. Centrosome, Nucleosome and Centrioles

48. The solid linear cytoskeleton elements having a diameter of 6 nm and made up of a single type of monomer are known as: (2014)

a. Lamins b. Microtubules
c. Microfilaments d. Intermediate filaments

49. Match the following and select the correct answer: (2014)

A.	Centriole	i.	Infoldings in mitochondria
B.	Chlorophyll	ii.	Thylakoids
C.	Cristae	iii.	Nucleic acids
D.	Ribozymes	iv.	Basal body cilia or flagella

- a. A-iv B-iii C-i D-ii
 b. A-iv B-ii C-i D-iii
 c. A-i B-ii C-iv D-iii
 d. A-i B-iii C-ii D-iv

Nucleus and Microbodies

50. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Fleming	(I)	Disc shaped sacs or cisternae near cell nucleus
(B)	Robert Brown	(II)	Chromatin
(C)	George Palade	(III)	Ribosomes
(D)	Camillo Golgi	(IV)	Nucleus

Choose the **correct** answer from the options given below:

- a. A-II, B-IV, C-III, D-I b. A-II, B-III, C-I, D-IV
 c. A-I, B-II, C-III, D-IV d. A-IV, B-II, C-III, D-I

51. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Metacentric chromosome	(I)	Chromosome has a terminal centromere
(B)	Sub-metacentric chromosome	(II)	Middle centromere forming two equal arms of chromosome
(C)	Acrocentric chromosome	(III)	Centromere is slightly away from the middle of chromosome resulting into two unequal arms
(D)	Telocentric chromosome	(IV)	Centromere is situated close to its end forming one extremely short and one very long arm

Choose the **correct** answer from the options given below:

- a. A-II, B-I, C-IV, D-III b. A-IV, B-I, C-II, D-III
 c. A-I, B-II, C-III, D-IV d. A-II, B-III, C-IV, D-I

52. Match List-I with List-II. (2024 Re)

List-I		List-II	
(A)	F ₁ Particles	(I)	Chromosomes
(B)	Histones	(II)	Cilia
(C)	Axoneme	(III)	Golgi apparatus
(D)	Cisternae	(IV)	Mitochondria

Choose the **correct** answer from the options given below:

- a. A-II, B-I, C-IV, D-III b. A-IV, B-I, C-II, D-III
 c. A-IV, B-I, C-III, D-II d. A-IV, B-III, C-I, D-II

53. Match List-I with List-II. (2024)

List-I		List-II	
(A)	Nucleolus	(I)	Site of formation of glycolipid
(B)	Centriole	(II)	Organization like the cartwheel
(C)	Leucoplasts	(III)	Site for active ribosomal RNA synthesis
(D)	Golgi apparatus	(IV)	For storing nutrients

Choose the correct answer from the options given below:

- a. A-(III), B-(IV), C-(II), D-(I)
 b. A-(I), B-(II), C-(III), D-(IV)
 c. A-(III), B-(II), C-(IV), D-(I)
 d. A-(II), B-(III), C-(I), D-(IV)

54. Match List-I with List-II: (2024)

List-I		List-II	
(A)	Axoneme	(I)	Centriole
(B)	Cartwheel pattern	(II)	Cilia and flagella
(C)	Crista	(III)	Chromosome
(D)	Satellite	(IV)	Mitochondria

Choose the correct answer from the options given below :

- a. A-(II), B-(IV), C-(I), D-(III)
 b. A-(II), B-(I), C-(IV), D-(III)
 c. A-(IV), B-(III), C-(II), D-(I)
 d. A-(IV), B-(II), C-(III), D-(I)

55. Match List-I with List-II. (2022)

List-I		List-II	
A.	Metacentric chromosome	i.	Centromere situated close to the end forming one extremely short and one very long arms
B.	Acrocentric chromosome	ii.	Centromere at the terminal end
C.	Sub-metacentric	iii.	Centromere in the middle forming two equal arms of chromosomes
D.	Telocentric chromosome	iv.	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the correct answer from the options given below.

- a. A-i B-ii C-iii D-iv
 b. A-iii B-i C-iv D-ii
 c. A-i B-iii C-ii D-iv
 d. A-ii B-iii C-iv D-i

56. Which of the following is an incorrect statement? (2021)

- a. Microbodies are present both in plant and animal cells.
 b. The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
 c. Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
 d. Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.

57. Match List-I with List-II (2021)

List-I		List-II	
A.	Cristae	i.	Primary constriction in chromosome
B.	Thylakoids	ii.	Disc-shaped sacs in Golgi apparatus
C.	Centromere	iii.	Infoldings in mitochondria
D.	Cisternae	iv.	Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below.

- a. A-i B-iv C-iii D-ii
 b. A-iii B-iv C-i D-ii
 c. A-ii B-iii C-iv D-i
 d. A-iv B-iii C-ii D-i

58. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as: (2021)

- a. Telocentric b. Sub-metacentric
 c. Acrocentric d. Metacentric

59. The biosynthesis of ribosomal RNA occurs in: (2020-Covid)

- a. Golgi apparatus b. Microbodies
 c. Nucleolus d. Ribosomes

60. The shorter and longer arms of a submetacentric chromosome are referred to as (2019)

- a. s-arm and l-arm respectively
 b. p-arm and q-arm respectively
 c. q-arm and p-arm respectively
 d. m-arm and n-arm respectively

61. Which of the following is true for nucleolus? (2018)

- a. Larger nucleoli are present in dividing cells.
 b. It is a membrane-bound structure.

c. It takes part in spindle formation.

d. It is a site for active ribosomal RNA synthesis

62. The chromosomes in which centromere are situated close to one end are: (2015)

- a. Telocentric b. Sub-metacentric
 c. Metacentric d. Acrocentric

63. Nuclear envelope is a derivative of: (2015)

- a. Microtubules
 b. Rough endoplasmic reticulum
 c. Smooth endoplasmic reticulum
 d. Membrane of Golgi complex

64. Balbiani rings are sites of: (2015 Re)

- a. Nucleotide synthesis
 b. Polysaccharide synthesis
 c. RNA and protein synthesis
 d. Lipid synthesis

65. Match the columns and identify the correct option. (2015 Re)

Column-I		Column-II	
A.	Thylakoids	i.	Disc-shaped sacs in Golgi apparatus
B.	Cristae	ii.	Condensed structure of DNA
C.	Cisternae	iii.	Flat membranous sacs in stroma
D.	Chromatin	iv.	Infoldings in mitochondria

- a. A-(iii), B-(iv), C-(i), D-(ii)
 b. A-(iii), B-(i), C-(iv), D-(ii)
 c. A-(iii), B-(iv), C-(ii), D-(i)
 d. A-(iv), B-(iii), C-(i), D-(ii)

Answer Key

1. (a) 2. (c) 3. (d) 4. (a) 5. (c) 6. (c) 7. (d) 8. (a) 9. (a) 10. (c)
 11. (a) 12. (a) 13. (d) 14. (d) 15. (d) 16. (a) 17. (b) 18. (d) 19. (a) 20. (c)
 21. (d) 22. (c) 23. (b) 24. (c) 25. (a) 26. (b) 27. (d) 28. (b) 29. (d) 30. (c)
 31. (d) 32. (c) 33. (a) 34. (c) 35. (a) 36. (d) 37. (a) 38. (c) 39. (b) 40. (d)
 41. (c) 42. (d) 43. (d) 44. (a) 45. (d) 46. (d) 47. (b) 48. (c) 49. (b) 50. (a)
 51. (d) 52. (b) 53. (c) 54. (b) 55. (b) 56. (d) 57. (b) 58. (d) 59. (c) 60. (b)
 61. (d) 62. (d) 63. (b) 64. (c) 65. (a)